

Jared Reichle

SUMMARY

Software engineer with a strong foundation in algorithms, data processing, and large-scale system reliability. I've spent my career building automation frameworks, integrating distributed systems, and shipping production tools that speed up development and testing. I'm currently pursuing an M.S. in Computer Science with an AI specialization, where I'm gaining hands-on experience with machine learning, deep learning, and statistical modeling. I enjoy taking ambiguous problems back to first principles, prototyping quickly, and working closely with engineering and research teams to build next-generation AI applications.

CORE SKILLS

Programming: Python, C++, C, TypeScript, JavaScript, Bash, PowerShell

ML & Data: PyTorch, scikit-learn, NumPy, Pandas, data processing, model evaluation

Backend & Systems Engineering: API development, distributed systems, data pipelines, automation frameworks, system design

DevOps & Infra: CI/CD pipelines, docker/containers, Ansible, Git, Azure, Linux (Debian/Ubuntu/RHEL)

Embedded & OS: VxWorks, low-level hardware integration, device drivers, real-time systems

Frontend: React, Node.js, Next.js

Databases: PostgreSQL, SQLite, SQLAlchemy

Soft Skills: Cross-team collaboration, mentoring, technical communication, requirements translation, product ownership

EDUCATION

Georgia Institute of Technology - Online M.S. Computer Science	January 2025 - Ongoing
- Specialization: Artificial Intelligence	

University of Idaho - B.S. Electrical Engineering	August 2016 - May 2020
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EMPLOYMENT

Principal Software Engineer	Northrop Grumman	August 2022 - Present
Sentinel LGM-35A Test Automation & Systems Engineering		
- Built test tools and automation frameworks from scratch, accelerating system test cycles - Created API clients and device drivers that integrated web tools with lab hardware - Developed a React app for interactive lab visualization, improving team situational awareness - Established DevOps infra with CI/CD pipelines, enhancing software delivery speed and reliability - Mentored new engineers, reducing onboarding time and improving team velocity by refining documentation and workflows		

Electronics Engineer	United States Air Force	July 2020 - July 2022
F-16 Fighting Falcon Software Maintenance		
- Managed customer-driven software changes, resolving bugs and adding features to F-16 subsystems that improved mission-readiness - Integrated cross-team feature updates into flight simulators, accelerating validation and reducing test cycles - Identified, documented and hotfixed simulator issues, decreasing field defects and improving system reliability		

Systems Engineer Debugging Memory Products

- Debugged memory product failures on proprietary customer systems, increasing first-call resolution rates
- Produced technical reports and directly interfaced with customers to drive faster problem closure and improved product trust

PROJECTS

Home Lab Servers

- Built home lab with Proxmox cluster on surplus hardware
- Set up VMs and containers to explore OS and networking.

FPGA Data Acquisition and Control System

- Created a modular and extendable FPGA data acquisition and control system for a laser interferometer closed loop system.
- Developed in a team with SystemVerilog to simulate designed feedback loop on Zynq 7010 SoC.
- Link: http://mindworks.shoutwiki.com/wiki/FPGA_Data_Acquisition_and_Control

Circuit Synthesis Algorithms

- Created a python project to take an N-input/output 'black-box' circuit characterized by its sweeping frequency response data set and convert them into RLC circuit branches.
- Worked independently over the course of a semester.

Stellarium Telescope Control Scripts

- Made a TCP listener that works with Stellarium (Free astronomy software) to listen for slew commands and translate that into Azimuth/Altitude encoded positions over RS-232 port.

ML& AI COURSEWORK PROJECTS (OMSCS)

- Implemented ML algorithms (regression, classification, random forests, KNN, bagging and boosting) in Python using scikit-learn and NumPy.
- Built reinforcement learning agents (tabular + function approximation)
- Developed neural network models in PyTorch for supervised tasks and model evaluation.
- Applied statistical methods to analyze datasets (like stock pricing data), evaluate model performance, and perform inference.

CERTIFICATIONS AND STATUSES

CompTia Security+ (SY0-701)

Date: 10/2025

Active Top Secret Clearance

Date: 06/2023